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ITC 413: NoSQL Databases

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Team Project Reflection: IoT Sensor Management System

For the ITC 413 team project, our group designed an IoT Sensor Management System using Cassandra through DataStax Astra DB. Because our team only had two members, we reduced the original project scope to focus on the core database requirements. The final implementation used three main entities: Sensor Readings, Device Registry, and Daily Aggregates. The project helped me better understand Cassandra database design, team coordination and personal accountability during a group project.

Technical Skills Development

The main technical skill I developed was Cassandra data modeling. This project made it clear that Cassandra design is different from relational database design. Instead of starting with normalized tables and planning joins, Cassandra design starts with the queries the system needs to support. Each table was built around a specific access pattern. For example, the `sensor_readings_by_sensor_day` table was designed to retrieve readings for one sensor on one day or within a time range. The partition key used `sensor_id` and `reading_date`, while `reading_timestamp` was used as a clustering column to keep readings ordered by time. This helped me understand why time-series data needs proper bucketing. I also learned more about denormalization. We created separate query-focused tables for threshold violations, latest readings by region, device status by region and daily aggregates. This duplicated some data, but

it made the queries more efficient. Before this project, I mostly thought of duplicate data as something to avoid. Now I understand that in Cassandra, denormalization is often required for fast and predictable queries.

Planning and Task Management

Our planning focused on reducing the project to something realistic for a two-person team. We kept Sensor Readings, Device Registry and Daily Aggregates, while cutting larger features such as full alert workflows, predictive maintenance, anomaly detection and compliance reporting. This was the right decision because it let us complete a working database instead of trying to build too much. Task management was not perfect. I ended up taking the lead on most of the database implementation because of time pressure and because I had another major project to focus on. Thomas helped with final documentation, database access setup and review. The final work distribution was not ideal, but we adapted and completed the required pieces. If I could change our process, I would set earlier internal deadlines. Having separate deadlines for schema design, sample data and query testing would have made the project easier to divide.

Communication and Coordination

Communication was respectful and practical. We coordinated through shared documents and messages, especially near the end when we had to finish the completion form, screenshots, and instructor access. One positive part of the project was that we avoided unnecessary conflict. Thomas acknowledged that I had done a large part of the implementation and I understood that he was also dealing with other coursework. Instead of making that a problem, we focused on finishing the project. He helped by setting up instructor access and contributing to the final submission document.

Personal Responsibility and Accountability

My main responsibility was the technical implementation. I helped design the Cassandra tables, create the query patterns, load sample data and prepare the database evidence for submission. The final queries supported time-range retrieval, threshold violation lookup, latest readings by region, daily aggregation reports and communication failure/status lookup. The database included threshold flags, a separate violations table, device statuses such as ACTIVE, OFFLINE, and MAINTENANCE and missing reading counts in the aggregate table. I made sure my assigned work was completed by pushing the project forward as the deadline got closer. At the same time, I learned that moving too quickly can make collaboration harder. In the future, I would still take initiative, but I would divide tasks earlier and communicate implementation plans more clearly.

This project helped me better understand Cassandra and NoSQL database design. I learned how to design around query patterns, use partition keys and clustering columns, bucket time-series data and create denormalized tables for efficient reads. I also became more comfortable using DataStax Astra DB. The team process was not perfect, but we managed the deadline and completed a working project. The biggest lessons for me were to reduce scope realistically, communicate earlier and set clearer internal checkpoints. Even with the challenges, the final project demonstrated the core Cassandra concepts required for the assignment and gave me useful experience working in a technical team environment.